

EXPERIENCE

VEHC Lab, Indian Institute of Technology Madras

Mechanical Engineering Intern

Chennai, Tamilnadu

May'24 - Jul'24

- Investigated the **nonlinear dynamics of a vertical cantilever beam** under inclined base excitation for **piezoelectric energy harvesting**, developing and analyzing the model using **Wolfram Mathematica, MATLAB, and Simulink**.

M.TECH THESIS

Control of Underactuated Bipedal Locomotion with Ankle Push-Off

May'26 - Jul'27

Guide: Prof. Arpita Sinha, Systems and Control, IIT Bombay

Co-Guide: Prof. Vivek Sangwan, Dept. of Mechanical Engg., IIT Bombay

- Developed an **underactuated planar biped** framework with **pneumatic push-off**, implementing **PID and MPC** for desired step sizes and stable walking; modified the hardware and characterized the **mechanical and motor dynamics** using **dSPACE**.
- Designing a **kinematic foot mechanism** and continuous-time control model, with planned extensions to **knee joints, MPC-based trajectory generation, vision-based control, and VLA models**.

PROJECTS

Receding Horizon Control for Autonomous Crazyflie Navigation | Course Project

Jan'26 - Apr'26

Course: SC 627, Motion Planning & Coordination of Autonomous Vehicles | Prof. Arpita Sinha

- Formulated **Crazyflie navigation** as a double-integrator optimal control problem, generated **collision-free trajectories**, and developed an **MPC-based Receding Horizon Controller** validated on hardware using **Qualisys motion capture**.

Autonomous Mobile Robot Motion Planning and Navigation | Course Project

Jan'26 - Apr'26

Course: SC 627, Motion Planning & Coordination of Autonomous Vehicles | Prof. Arpita Sinha

- Generated **2D occupancy maps** using SLAM, formulated the robot **C-space**, and implemented **A* path planning from scratch**; integrated **AMCL and costmap tuning** and validated navigation on a physical **TurtleBot**.

Deep Reinforcement Learning for Robot Locomotion | Course Project

Jan'26 - Apr'26

Course: IE 620, Reinforcement Learning Algorithms | Prof. Nandyala Hemachandra

- Reproduced a **Deep Reinforcement Learning** control approach using custom **PyBullet/Gymnasium** environments and implemented the **ADDPG actor-critic framework from scratch** for autonomous robotic control.

Computed Torque Control with RPP for Robotic Manipulators | Research Project

Jan'25 - Aug'25

In collaboration with Prof. Anirudh Nath and Prof. Anirban Nag, IIST Shibpur

- Designed **Computed Torque Control** for **2-DOF and 6-DOF manipulators** with nonlinear dynamics and **LMI-based Regional Pole Placement**, evaluating trajectory tracking and robustness using **MATLAB/Simulink**.

TECHNICAL SKILLS

Programming: Python, C++, MATLAB | **Control & Modeling:** Control Systems, Simulink, Simscape, dSPACE

Robotics: ROS/ROS2, Gazebo, Webots, MuJoCo | **Hardware:** STM32, Arduino, ESP32, Jetson Nano, Raspberry Pi

ACHIEVEMENTS

- Departmental Topper** and recipient of the **Gold Medal** for B.Tech. in Mechanical Engineering.
- Published **Nonlinear Dynamics of a Vertical Cantilever Beam Under Biaxial Base Excitation** in the *Journal of Computational and Nonlinear Dynamics*.
- Co-authored papers published in **IEEE Xplore** and as a **Springer book chapter** on robotic control and thermoelectric generators.

POSITION OF RESPONSIBILITY

IMechE Student Chapter | Vice Chairman, IIST Shibpur

Jul'24 - Jul'25

- Led **technical, outreach, and industrial initiatives** including CAD 2.0, star-gazing sessions, industrial visits, and served as **Organizing Lead for the SOFE Eastern Region Finals (Oct 2023)**.